

But the period and radius of orbit have unique values.

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Total initial energy of the satellite:

$$\begin{aligned} E_i &= GPE_i + KE_i = -\frac{GMm}{r_i} + \frac{1}{2}mv_i^2 \\ &= -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(690)}{7.2 \times 10^6} + \frac{1}{2}(690)(7.5 \times 10^3)^2 \\ &= -1.895 \times 10^{10} \text{ J} \end{aligned}$$

Total final energy of the satellite:

$$\begin{aligned} E_f &= GPE_f + KE_f = -\frac{GMm}{r_f} + \frac{1}{2}mv_f^2 \\ &= -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(690)}{6.5 \times 10^6} + \frac{1}{2}(690)(7.9 \times 10^3)^2 \\ &= -2.095 \times 10^{10} \text{ J} \end{aligned}$$

$E_f < E_i$, so the satellite loses total energy.

Change in total energy = $E_f - E_i = -2.0 \times 10^9 \text{ J}$

In other words, the satellite loses approximately $2 \times 10^9 \text{ J}$.

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